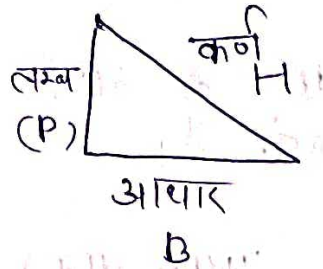


Basic Trigonometry

Trigonometry त्रिकोणमिति



$$\sin \theta = \frac{\text{लम्ब}}{\text{कर्ण}} = \frac{P}{H}$$

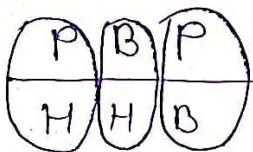
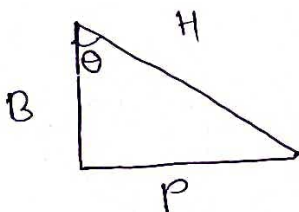
$$\operatorname{Cosec} \theta = \frac{1}{\sin \theta} = \frac{\text{कर्ण}}{\text{लम्ब}} = \frac{H}{P}$$

$$\cos \theta = \frac{\text{आधार}}{\text{कर्ण}} = \frac{B}{H}$$

$$\sec \theta = \frac{1}{\cos \theta} = \frac{\text{कर्ण}}{\text{आधार}} = \frac{H}{B}$$

$$\tan \theta = \frac{\text{लम्ब}}{\text{आधार}} = \frac{P}{B}$$

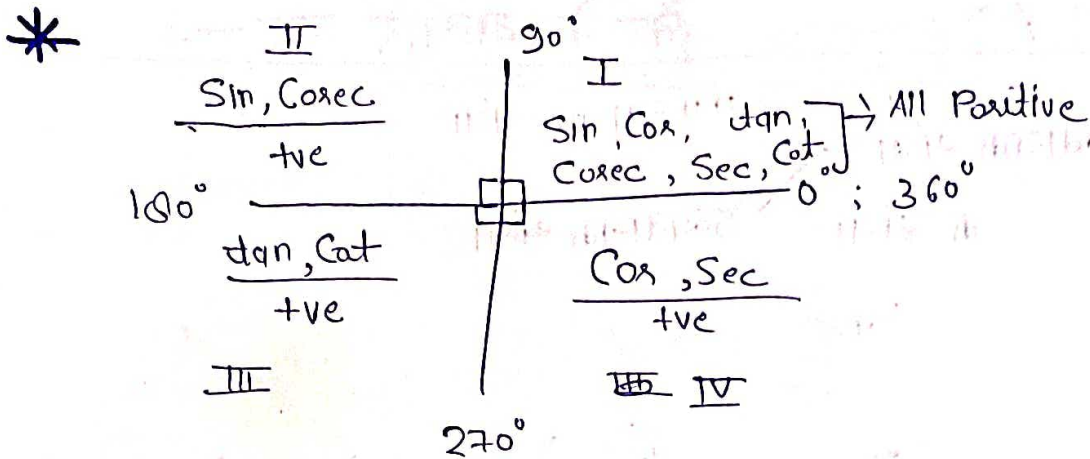
$$\cot \theta = \frac{1}{\tan \theta} = \frac{\text{आधार}}{\text{लम्ब}} = \frac{B}{P}$$



Sin Cos Tan



Sin Cos Tan



* $\boxed{90 + \theta}$

$$\sin(90 + \theta) = +\cos \theta$$

$$\cos(90 + \theta) = -\sin \theta$$

$$\tan(90 + \theta) = -\cot \theta$$

$$\operatorname{cosec}(90 + \theta) = +\sec \theta$$

$$\sec(90 + \theta) = -\operatorname{cosec} \theta$$

$$\cot(90 + \theta) = -\tan \theta$$

* $\boxed{270 + \theta}$

$$\sin(270 + \theta) = -\cos \theta$$

$$\cos(270 + \theta) = +\sin \theta$$

et

$$\tan (270 + \theta) = -\cot \theta$$

$$\operatorname{Cosec} (270 + \theta) = -\sec \theta$$

$$\sec (270 + \theta) = +\operatorname{cosec} \theta$$

$$\cot (270 + \theta) = -\tan \theta$$

* $(180 + \theta)$ No change

$$\sin (180 + \theta) = -\sin \theta$$

$$\cos (180 + \theta) = -\cos \theta$$

$$\tan (180 + \theta) = +\tan \theta$$

$$\operatorname{Cosec} (180 + \theta) = -\operatorname{cosec} \theta$$

$$\sec (180 + \theta) = -\sec \theta$$

$$\cot (180 + \theta) = +\cot \theta$$

Question $\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ$ का मान है

$$\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{1}{2}$$

$$\frac{3}{4} - \frac{1}{4}$$

$$\frac{2}{4} = \frac{1}{2} \text{ Ans}$$



बिहार पुलिस भर्ती

MATHS



बुद्धा वैद्य



Angle (In Degrees)	0°	30°	45°	60°	90°	180°	270°	360°
Angle (In Radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not Defined	0	Not Defined	-1
cot	Not Defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	Not Defined	0	Not Defined
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not Defined	-1	Not Defined	1
cosec	Not Defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	Not Defined	-1	Not Defined

Question $\sin^2 30^\circ, \cos^2 45^\circ + 4 \tan^2 30^\circ + (\sin^2 90^\circ) /$
 $2 + 2 \cos 90^\circ$ का मान है

$$\left(\frac{1}{2}\right)^2 \times \left(\frac{1}{\sqrt{2}}\right)^2 + 4 \times \left(\frac{1}{\sqrt{3}}\right)^2 + \frac{(1)^2}{2} + 0$$

$$\frac{1}{4} \times \frac{1}{2} + 4 \times \frac{1}{3} + \frac{1}{2}$$

$$\frac{1}{8} + \frac{4}{3} + \frac{1}{2}$$

$$\frac{35}{24} + \frac{1}{2} \rightarrow \frac{35}{24} + \frac{12}{24} = \frac{47}{24} \text{ Ans}$$

Question - यदि $\sin \theta = 1/2$ तो $\frac{\sin^3 \theta + \cos^2 \theta + \tan^2 \theta}{\cos^2 \theta + \operatorname{cosec} \theta - \sin \theta}$
 का मान ज्ञात कीजिए

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\frac{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{\sqrt{3}}\right)^2}{\left(\frac{\sqrt{3}}{2}\right)^2 + 2 - \frac{1}{2}}$$

$$\frac{\frac{1}{4} + \frac{3}{4} + \frac{1}{3}}{\frac{3}{4} + \frac{3}{2}} = \frac{1 + \frac{1}{3}}{\frac{10}{3}} = \frac{\frac{4}{3}}{\frac{10}{3}}$$

$$\frac{4 \times 3}{2 \times 10} = \frac{16}{27} \text{ Ans}$$

Question $\cos \theta = 1/2$ और $P = \tan \theta$, यदि $0^\circ < \theta < 90^\circ$ है तो $P^2 - SP - 2$ का मान ज्ञात कीजिए

$$\theta = 60^\circ$$

$$P = \tan 60^\circ$$

$$P = \sqrt{3}$$

$$P^2 - SP - 2$$

$$(\sqrt{3})^2 - S \times \sqrt{3} - 2$$

$$3 - S \sqrt{3} - 2$$

$$1 - S \sqrt{3} \text{ Ans}$$